Cost, Carbon, and the Speed of Replenishment in Distribution Network Design

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Abstract

We develop a quantitative decision model for carbon-induced operational substitution in distribution network design. A firm distributes a single product from one source through a network of distribution centers to many markets and chooses replenishment speed and network density to minimize cost subject to a carbon cap and a service requirement. Distribution carbon arises from moving goods, operating facilities, and holding inventory. The model treats availability and proximity as services that can each be provided by two activities: availability through safety stock or replenishment speed, and proximity through facilities or last-mile travel. Carbon accounting can select a different activity than cost minimization because each substitution reduces one source of emissions while increasing another. Slower replenishment reduces freight emissions but increases safety stock; consolidation reduces facility carbon and improves risk pooling but lengthens the last mile. We derive closed-form thresholds that identify the carbon-minimizing direction of each decision for a given product. The speed threshold compares holding carbon per dollar with inbound-transport carbon per dollar. The density threshold compares last-mile carbon per dollar with a pooling-adjusted combination of facility and holding intensities. Speed and density are complements, so the final configuration follows from the joint decision. A chance-constrained cap on realized carbon shifts both thresholds through a throughput-risk multiplier. Numerical illustrations show how the thresholds classify product archetypes and support activity-level carbon accounting.

Keywords: Logistics; Distribution network design; Carbon emissions; Safety stock; Replenishment speed

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1 Introduction

A firm serves its markets from a network of distribution centers that it replenishes over time. Among the operational choices this involves, we study two: the *density* of the network — how many distribution centers the firm operates — and the *speed* with which those centers are replenished. Both shape the firm's cost, and both shape the carbon its distribution emits. As carbon is increasingly priced and reported (McKinnon and Piecyk 2012), a firm sets both with carbon as well as cost in view, and we study how a carbon price shapes that choice.

Each decision carries a standard cost–carbon trade-off. Network density governs proximity to markets: operating more distribution centers shortens the last-mile distance and changes the warehouse energy the network consumes. Replenishment speed governs availability: replenishing faster shortens the lead time that inventory must cover and changes the emissions of the freight that serves the network. Distribution emits carbon both when goods move and when they are held, and each affects both the firm's cost and its carbon. Cost balances these effects one way; with a price on carbon, the carbon-minimizing balance can fall elsewhere, and how it does so is the central question this paper addresses.

Slowing replenishment and consolidating the network are the common transport-side responses: slowing moves freight by slower, lower-emission modes, as in slow steaming in shipping (Psaraftis and Kontovas 2013; Cariou 2011), and consolidating serves markets from fewer, larger distribution centers (McKinnon and Piecyk 2012). Each response also affects inventory. A unit held in a warehouse consumes energy — refrigeration, climate control, handling — for as long as it is held. Slowing replenishment lengthens the lead time that a stock buffer must cover, so the safety stock, and the carbon of holding it, rises with the square root of that lead time; consolidating pools demand across markets (Eppen 1979) and saves facility energy while lengthening the last mile. Each substitution thus lowers one source of carbon and raises another, and whether it lowers the total depends on which source is larger.

The choice between the two sources is a choice between two operational substitutions. A firm provides *availability* either by holding safety stock or by replenishing faster — a shorter lead time leaves less for the buffer to cover, so speed and inventory are substitutes for the same service — and it provides market *proximity* either by operating more local facilities or by delivering farther from fewer, so facilities and last-mile travel are substitutes for the same access. The cost-minimizing configuration uses one mix of these activities; when carbon is priced, the optimal mix can differ. When safety stock is more carbon-intensive per dollar than the inbound transport that would replace it, the carbon-minimizing chain provides availability through *faster* replenishment. When last-mile travel is more carbon-intensive per dollar than the facilities that would shorten it, the carbon-minimizing chain provides proximity through facilities — a *more dispersed* network. The lowest-carbon distribution system thus depends on the product: a cold-chain grocer and a bulk dry-goods shipper have different optimal configurations.

We formalize this in a single-product location–inventory model with stochastic demand. The firm chooses network density and replenishment speed to minimize cost subject to a carbon cap

and a service-level requirement. Cost and carbon each comprise the same four components — facilities, holding, inbound transport, and last-mile transport. The model follows the location–inventory tradition (Daskin et al. 2002; Shen et al. 2003) in a continuous-approximation form (Daganzo 2005), with a distinguishing addition: replenishment speed is a decision. Because speed sets the lead time that stochastic safety stock must cover, it directly determines inventory carbon and couples the four components together.

The paper contributes a threshold model for this joint decision. The organizing quantity is the *marginal carbon intensity per dollar* of each activity, and the model yields three structural results. First, a *speed reversal* in the availability substitution: holding the network fixed, total carbon is U-shaped in speed, and the carbon-minimizing speed exceeds the cost-minimizing one when holding is more carbon-intensive per dollar than inbound transport at the margin, so a carbon constraint can raise the optimal replenishment speed — a distinct configuration from the slow-steaming literature. Second, a *pooling-adjusted density threshold* in the proximity substitution. The carbon-minimizing network is more dispersed when the last mile is carbon-intensive per dollar relative to facilities. Adding a facility, however, also gives up some demand pooling, so the threshold is not that facility comparison alone: it blends in the holding intensity, and therefore depends on demand correlation, service, and volatility. Third, the two decisions are *complements*: faster replenishment shrinks safety stock and lowers the inventory penalty of dispersing, so the carbon-minimizing response adjusts both together rather than one alone. Together, the two thresholds give a product-specific diagnostic for the joint design problem: each compares the carbon intensity per dollar of the two competing activities and identifies the carbon-minimizing direction — faster or slower replenishment, a denser or more dispersed network — placing a product in one of four regimes. Scientifically, the thresholds characterize when carbon accounting changes the operational activity selected by an optimal distribution design. Practically, they require activity-level cost and carbon rates that firms can estimate from logistics data. In one regime, both comparisons point to a network that is at once faster *and* more dispersed. A chance-constrained extension, in which the cap bounds realized rather than expected carbon, tightens the speed threshold and relaxes the density one.

The remainder of the paper proceeds as follows. Section 2 positions the paper relative to the location–inventory, carbon-network-design, and inventory–transport–emissions literatures. Section 3 develops the model, Section 4 establishes the results, Section 5 reports numerical illustrations, Section 6 draws out the managerial implications, and Section 7 concludes.

2 Related Literature

This study connects four streams: location–inventory design with risk pooling, carbon-aware network design, the joint control of inventory and transport mode under emissions regulation, and carbon as a priced operational cost in service-constrained inventory systems. Each develops an element of the two substitutions examined here — availability (inventory versus replenishment speed)

and proximity (facilities versus last-mile travel). Each of these streams is well developed, and we draw on all four; we position the contribution by what it adds to each: the joint design of network density and replenishment speed under a carbon cap and a service requirement, together with the closed-form thresholds that relate the carbon-minimizing configuration to the cost-minimizing one.

2.1 Location–inventory design and risk pooling

A first stream establishes how demand risk pooling shapes distribution network design: jointly choosing facility location and safety stock, it characterizes the cost-optimal number and placement of distribution centers. Eppen (1979) characterizes the effect of centralization on expected cost in the multi-location newsvendor, establishing the square-root relationship between the number of stocking locations and pooled safety stock. Daskin et al. (2002) and Shen et al. (2003) embed working and safety stock into the facility-location decision, so that risk pooling shapes the optimal number and placement of distribution centers, with extensions to stochastic settings by Snyder et al. (2007), to capacitated warehouses by Özsen et al. (2008), and to continuous-approximation allocation by Tsao et al. (2012). Daganzo (2005) provides the continuous-approximation foundation in which last-mile distance scales with the inverse square root of the number of facilities. Closest to our spatial–inventory coupling, Straubert (2024) ties the number of stocking locations to an exact reorder-point/order-quantity inventory model through a delivery-lead-time penalty, solved by continuous approximation. The centralize-versus-decentralize question these models pose can also turn on forces beyond demand pooling: Schmitt et al. (2015) show analytically that supply-disruption risk diversification can favor decentralization for a risk-averse firm, in contrast to the classical pooling result. These works develop network density and safety-stock placement under risk pooling, with a cost or service objective and no carbon; they do not make replenishment speed a continuous decision coupled with carbon. The present study retains the risk-pooling structure and the density decision and adds an endogenous replenishment speed together with a carbon objective.

2.2 Carbon-aware network design

A second stream brings carbon into facility location and network design, characterizing how an emissions objective or cap can change the optimal network configuration. Cachon (2014) relates retail store density to greenhouse gas cost, with consumer travel to stores as the dominant emission channel and inventory entering through retail floor space. Turken et al. (2017) examine facility location and capacity under a carbon tax and emission limits, weighing centralization against decentralization through production scale economies; Harris et al. (2011) compare cost-optimal and emissions-aware network designs numerically; Wang et al. (2020) develop a carbon-compliant location–inventory model under demand and carbon-price uncertainty; Li et al. (2017) jointly optimize distribution-network design and two-echelon inventory under a CO_2 emission tax; and Hacardiaux and Tancrez (2020) assess the transport-emission savings of a shared, consolidated location–inventory network under horizontal cooperation, with Hacardiaux et al. (2022) balancing partner preferences over logistics cost and carbon footprint in the same cooperative setting. These studies

make density or facility configuration the central decision and bring emissions into the objective, the constraints, or an ex post assessment; where transport is modeled as an operational choice, it enters through travel distances, discrete modes, or lead-time parameters rather than through a continuously optimized replenishment speed chosen jointly with density. The present study treats network density and replenishment speed as joint decisions and resolves the proximity substitution between facilities and last-mile travel analytically.

2.3 Inventory, transport mode, and emissions

A third stream couples transport-mode and lead-time choices with inventory control under emissions regulation, characterizing how carbon reshapes the trade-off between freight and the inventory it serves. Hoen et al. (2014) and Chen and Wang (2016) study transport mode selection alongside inventory under stochastic demand and emissions regulation; Bektaş and Laporte (2011) formalize the load–emission relationship in the pollution-routing problem, embedding fuel use and emissions as a function of vehicle load in the routing objective; Moradi et al. (2025) optimize driving speed and path selection under load- and grade-dependent fuel use at the routing level; Konur and Schaefer (2014) solve the economic order quantity model with less-than-truckload and truckload carriers under carbon cap, tax, and cap-and-trade regulation, and Schaefer and Konur (2015) extend this to a continuous-review, reorder-point inventory system under stochastic demand, characterizing the cost–emission Pareto frontier and analyzing lead-time and demand-variance sensitivity. Drent et al. (2023) analyze dynamic dual supply-mode selection under an emission target, and Hammami et al. (2015) bring carbon from manufacturing, ordering, and inventory into a multi-echelon production–inventory model with lead-time constraints. Across these papers, transport service is represented through route/load physics, carrier class, discrete supply mode, or lead-time parameters and constraints; replenishment speed is not a continuous decision chosen jointly with network density. The present study adds to this stream the continuous speed–density threshold that results when replenishment speed and network density are chosen jointly under a carbon cap and a service requirement.

Boone and Ganeshan (2025) choose a discrete freight mode against inventory under an emission cap at a single location, with no network-density decision. The present study replaces the finite mode set with a continuous replenishment-speed decision, prices carbon on the holding as well as the transport side, embeds it in a network whose density is chosen jointly, and characterizes the resulting continuous speed–density threshold and complementarity.

2.4 Carbon policy, service, and safety-stock placement

A fourth stream prices carbon directly into inventory and service decisions, showing how a carbon price or cap shifts order quantities and safety-stock placement. Hua et al. (2011) and Bouchery et al. (2012) develop carbon-aware order-quantity models under cap-and-trade and multi-objective formulations, and Benjaafar et al. (2013) position cap, tax, and cap-and-trade as priced operational

instruments in simple supply-chain models; Chen et al. (2013) derive the closed-form carbon-constrained economic order quantity, the single-location analog of the marginal carbon-intensity logic we extend to a network, and Arıkan and Jammernegg (2014) impose a product carbon-footprint constraint on a dual-sourcing newsvendor. Silbermayr et al. (2017) bring environmental constraints into inventory pooling with dependent demand. Within multi-echelon service models, Ni and Shu (2015) study the trade-off between service time and carbon in safety-stock placement under the guaranteed-service framework, and Wang et al. (2025) incorporate cap-and-trade into guaranteed-service placement. Probabilistic carbon limits have been modeled through chance constraints in lot sizing (Koca 2021) and in supply-chain network design (Shaw et al. 2016), typically inside large mixed-integer programs; we instead obtain a closed-form deterministic equivalent that scales the transport component by a single factor. A recent review (Sebatjane 2025) surveys carbon-aware inventory models and the regulatory instruments they treat; across this body of work, transport-emission modeling, service and inventory, network design, and demand or carbon uncertainty have developed in largely separate streams, and we bring these strands together. The service-placement models in particular price carbon against service and safety stock, with the joint design of network density and replenishment speed outside their scope.

Prior work covers carbon-aware density, mode and lead-time choice, service-constrained placement, and chance constraints on carbon. The present study couples endogenous replenishment speed with network density under carbon and service constraints and derives marginal carbon-intensity thresholds for the joint design. The resulting diagnostic identifies which activity supplies availability and proximity when carbon is priced: a speed threshold comparing the carbon intensity per dollar of holding with that of inbound transport, a pooling-adjusted density threshold comparing last-mile intensity with a weighted average of facility and holding intensities, and the complementarity that links them.

3 The Model

3.1 Operational environment

Region and markets. A firm serves N markets distributed over a service region of area A. Per-period demand at market i is a random variable with mean μ and standard deviation σ, identically distributed across markets with common pairwise correlation $\rho \in [0, 1)$. Demand is stationary and the setting is single-product. Markets and distribution centers are treated symmetrically, the standard simplification that admits closed-form continuous-approximation results (Daganzo 2005).

Distribution centers and geography. The firm opens n identical distribution centers (DCs), $1 \le n \le N$, each serving a sub-region of area A/n. By the continuous-approximation spatial law, the mean last-mile (DC-to-market) distance falls as the square root of the served area,

$$d(n) \;=\; \delta\, n^{-1/2}, \qquad \delta \propto \sqrt{A}, \tag{1}$$

so a denser network shortens delivery but only at a decreasing rate (Daganzo 2005).

Replenishment and the speed decision. Each DC is replenished directly from the source at a chosen speed $v \in [v_{\min}, v_{\max}]$. Individual DCs sit at different distances from the source; what enters cost and carbon is total inbound freight: with identical DCs each carrying throughput $N\mu/n$, this is $\sum_j (N\mu/n)\, D_j = N\mu\, \bar{D}$, the total throughput times the *mean* source-to-DC distance $\bar{D}$. Because the DCs fill a region of fixed size served from a fixed source, $\bar{D}$ is set by the region's geometry (order $\sqrt{A}$) and is approximately invariant to n; we write $D \equiv \bar{D}$ and treat it as constant.[1] The replenishment lead time is then

$$L(v) \;=\; D/v. \tag{2}$$

Faster replenishment is simultaneously more expensive and more carbon-intensive. The maintained assumption is only that the inbound cost rate $t(v)$ and emission rate $\varepsilon(v)$ (both per unit per mile) are increasing and convex in v; for the closed-form results, we use the quadratic instantiation $\varepsilon(v) = \varepsilon_0 v^2$ and $t(v) = t_0 + t_1 v^2$, the canonical form motivated by the cubic fuel-speed law of slow steaming (Psaraftis and Kontovas 2013; Cariou 2011). We read v as a general index of replenishment speed — vessel speed, or equivalently a faster and more carbon-intensive mode — so slow steaming is a motivating special case, not the only interpretation. Treating replenishment speed as a continuously optimized decision, chosen jointly with network density, couples the transport and facility-and-holding components in this distribution setting.

Inventory and service. Each DC operates a base-stock policy and must meet a cycle service level α — the in-stock probability at the end of a period, the Type-1 (cycle) service measure. With normally distributed demand the safety factor is $z = \Phi^{-1}(\alpha)$, with $\alpha > \frac{1}{2}$ so that $z > 0$, taken here as a service requirement (a parameter). A DC that pools $g = N/n$ markets faces aggregate per-period demand with standard deviation $\sigma\sqrt{g\,[\,1 + (g-1)\rho\,]}$, the statistical economy of scale in inventory (Eppen 1979). Covering lead-time demand, each DC holds safety stock $z\,\sigma\sqrt{L}\,\sqrt{g[1 + (g-1)\rho]}$. Summing over the n DCs and writing the *pooling expression*

$$\Psi(n) \;=\; N\rho + n(1-\rho), \tag{3}$$

the system-wide safety stock is

$$\mathrm{SS}(n, v) \;=\; z\,\sigma\,\sqrt{D/v}\,\sqrt{N\,\Psi(n)}. \tag{4}$$

[1] Two modeling reductions are embedded here. First, the symmetric-DC reduction assigns every DC the representative lead time $L = \bar{D}/v$ in place of its own $L_j = D_j/v$. Safety stock enters through $\sqrt{L}$, which is concave in distance, so by Jensen's inequality $\sqrt{\bar{D}} \geq n^{-1}\sum_j \sqrt{D_j}$ (Jensen 1906): using the mean distance slightly *overstates* aggregate safety stock, by a gap that is second-order in the dispersion of the D_j and reduces as the distances equalize. Second, replenishment is point-to-point; were the DCs served on a consolidated route, the inbound distance would grow as $\sqrt{n}$ in the manner of the store-delivery tour of Cachon (2014), adding a term that reinforces centralization in Proposition 4 and leaving the fixed-density speed comparison of Proposition 2 unchanged; with inbound distance varying in n, however, the speed response would depend on density through $D(n)$, so the complementarity of Proposition 5 need not carry over unchanged.

At $\rho = 0$, (4) reduces to $z\sigma\sqrt{L}\sqrt{nN}$, the square-root law: decentralizing from 1 to N DCs multiplies safety stock by $\sqrt{N}$. As $\rho \to 1$, $\Psi(n) \to N$ and safety stock becomes invariant to n — the pooling benefit vanishes.

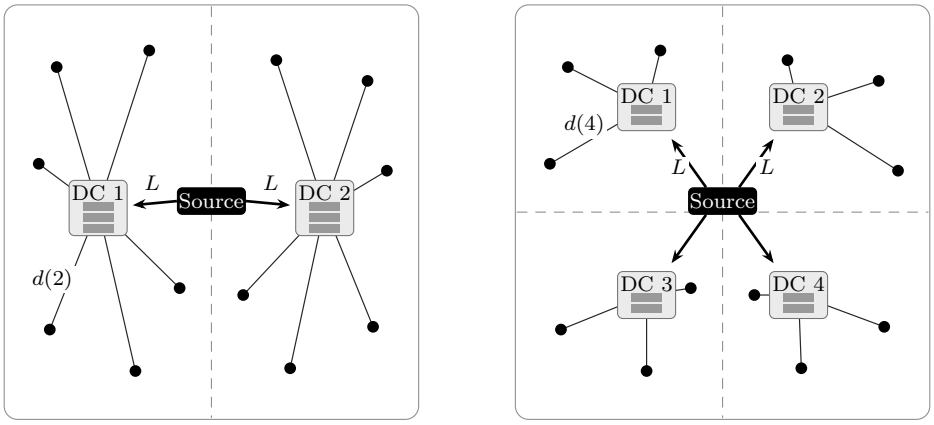
Decisions. The firm chooses the network density $n \in [1, N]$ (relaxed to a continuous variable and rounded) and the replenishment speed $v \in [v_{\min}, v_{\max}]$. Service is a requirement, so α (hence z) is a parameter in our model. Table 1 collects the notation.

3.2 Cost and carbon

Both the cost and carbon functions are system-wide flows over a single representative period, consistent with the stationary single-product environment above: F, φ are charged per DC per period, h, η per unit held per period, and the transport terms accrue against the per-period throughput. The representative distances in Figure 1 therefore enter global cost and carbon through aggregate freight tasks, not through a per-territory objective. Throughput itself is conserved at $N\mu$ regardless of the configuration: the firm moves the mean demand of all N markets through the network, whatever density n and speed v it chooses. With per-period throughput $N\mu$,

$$K(n, v) = \underbrace{F\, n}_{\text{fixed}} + \underbrace{h\, \mathrm{SS}(n, v)}_{\text{holding}} + \underbrace{t(v)\, N\mu D}_{\text{inbound}} + \underbrace{t_L\, N\mu\, d(n)}_{\text{last-mile}}, \tag{5}$$

$$C(n, v) = \underbrace{\varphi\, n}_{\text{fixed}} + \underbrace{\eta\, \mathrm{SS}(n, v)}_{\text{holding}} + \underbrace{\varepsilon(v)\, N\mu D}_{\text{inbound}} + \underbrace{\varepsilon_L\, N\mu\, d(n)}_{\text{last-mile}}, \tag{6}$$



$n = 2$: each DC serves 6 markets
more stock per DC, longer delivery legs

$n = 4$: each DC serves 3 markets
less stock per DC, shorter delivery legs

Faster replenishment (shorter L), at a given density:
system-wide safety stock shrinks, so holding cost and carbon fall, while freight cost and carbon rise.

More DCs (higher n), at a given speed:
delivery legs $d(n)$ shorten with territory area, so system-wide last-mile cost and carbon fall, while facility cost and carbon rise and weaker pooling raises system-wide holding cost and carbon.

Figure 1: Model representation for 12 markets (N) and 2 or 4 DCs ($n = 2$ or 4).

Table 1: Notation. The decision variables are the network density n and the replenishment speed v; all other quantities are exogenous parameters or derived.

Symbol	**Meaning**
Decision variables	
n	number of DCs (network density), $1 \le n \le N$
v	replenishment speed, $v \in [v_{\min}, v_{\max}]$
Demand (exogenous)	
N	number of markets
$\mu,\ \sigma$	mean, standard deviation of demand per market per period
ρ	cross-market demand correlation, $\rho \in [0, 1)$
Geography, cost, carbon (exogenous)	
$A,\ D,\ \delta$	region area; inbound distance; last-mile distance constant
$F,\ \varphi$	fixed cost, carbon per DC per period
$h,\ \eta$	holding cost, carbon per unit per period
$t(v),\ \varepsilon(v)$	inbound transport cost, carbon per unit-mile (convex in v)
$t_L,\ \varepsilon_L$	last-mile transport cost, carbon per unit-mile
α	cycle service level (in-stock prob.); $z = \Phi^{-1}(\alpha)$
$\tau,\ B$	carbon price; carbon cap
Derived	
$g = N/n$	markets served per DC
$L = D/v$	replenishment lead time
$d(n) = \delta\, n^{-1/2}$	mean last-mile distance
$\Psi(n) = N\rho + n(1-\rho)$	pooling expression
$\mathrm{SS}(n, v)$	system-wide safety stock, eq. (4)

where F, φ are fixed cost/carbon per DC, h, η are holding cost/carbon per unit per period, and t_L, ε_L are last-mile cost/carbon rates (Shen et al. 2003; Hua et al. 2011; Benjaafar et al. 2013). Each term multiplies a rate by the activity it prices; the two functions differ only in whether the rate is measured in dollars or kilograms.

Facilities. The facility terms Fn and φn are the periodic cost and carbon of operating the open DCs. They rise linearly in n and do not depend on v. Facility carbon φ is distinct from holding carbon η: φ captures baseline building energy and embodied carbon, whereas η prices the incremental carbon of storing a unit (for example refrigeration or handling).

Holding. The holding terms $h\mathrm{SS}(n, v)$ and $\eta\mathrm{SS}(n, v)$ value the system-wide safety stock of (4). They rise in n through the loss of pooling in $\Psi(n)$ and fall in v through the lead-time factor $\sqrt{D/v}$. This is the channel through which replenishment speed couples transport and inventory.

Inbound transport. The inbound terms multiply the representative freight task $N\mu D$ by the speed-dependent rates $t(v)$ and $\varepsilon(v)$. Since D is treated as invariant to n, inbound transport does not vary with network density; it rises in v through the increasing convex rates $t(v)$ and $\varepsilon(v)$, opposing the holding term that falls in v.

Last-mile transport. The last-mile terms multiply throughput $N\mu$ by the mean DC-to-market distance $d(n) = \delta n^{-1/2}$ and the rates t_L, ε_L. A denser network shortens this leg at the continuous-approximation rate, so the last-mile term is decreasing and convex in n and independent of v under constant per-mile last-mile rates.

Decomposition and scope. The carbon function splits into a transport component $\varepsilon(v)N\mu D + \varepsilon_L N\mu d(n)$ and a facility-and-holding component $\varphi n + \eta \mathrm{SS}(n, v)$; the cost function has the analogous partition. Slowing replenishment reduces inbound transport carbon but raises safety stock, while dispersion reduces last-mile distance but raises facility count and pooling loss. Three scope points follow. Cycle stock under a fixed review period is approximately $N\mu/2$ system-wide, or $(N\mu/n)/2$ per DC, and is invariant to both decisions. Pipeline inventory $N\mu D/v$ is omitted; when valued at the same dollar and carbon holding rates as stored inventory, it changes the level of v^* but not the speed threshold (Section 4). Production and procurement are also excluded because throughput is fixed at $N\mu$ and therefore constant in both decisions.

3.3 The program

The firm minimizes cost subject to a carbon cap B and the service requirement,

$$\min_{1 \le n \le N,\ v_{\min} \le v \le v_{\max}} K(n, v) \quad \text{s.t.} \quad C(n, v) \le B. \tag{7}$$

Here K and C are expected flows over the model's accounting period: under the stationary demand assumption they are the long-run average cost and carbon of a configuration, and they can be scaled to an annual expected-carbon cap by multiplying by the number of periods. Because demand is random, realized carbon over a specified accounting horizon fluctuates about its expectation; we develop the analysis for the expected-value cap and return below to a chance-constrained cap that bounds realized carbon with specified confidence (Proposition 7).

Equivalently, pricing carbon at $\tau \ge 0$ gives the priced objective $J = K + \tau C$. At an interior optimum, the cap multiplier equals the price ($\lambda = \tau$), as in carbon-constrained inventory models (Chen et al. 2013); sweeping τ traces the cost–carbon frontier within a stable interior regime, with jumps possible at regime switches or integer-density changes. Define the blended rates

$$F_\tau = F + \tau\varphi, \qquad H_\tau = h + \tau\eta, \qquad T_\tau(v) = t(v) + \tau\varepsilon(v), \qquad \ell_\tau = t_L + \tau\varepsilon_L.$$

Then

$$J(n, v; \tau) \;=\; F_\tau n + H_\tau \,\mathrm{SS}(n, v) + T_\tau(v) N\mu D + \ell_\tau N\mu d(n). \tag{8}$$

Setting $\tau = 0$ recovers cost; the limit $\tau \to \infty$ recovers carbon up to scale.

4 Analysis

We now characterize how carbon pricing changes the joint design. The joint first-order conditions decompose into two marginal substitutions: Proposition 2 gives the speed threshold on the availability margin, Proposition 3 records the service and volatility comparative static, Proposition 4 gives the pooling-adjusted density threshold on the proximity margin, Proposition 5 establishes the complementarity through which the two margins interact, and Corollary 6 characterizes the joint configuration when both comparisons agree. The results are analytical under the stated interiority and convexity conditions, each threshold derived on its margin with the other decision held at an arbitrary level; Section 5 evaluates the joint problem in calibrated numerical illustrations.

Lemma 1 (Carbon intensity per dollar)**.** *Pricing carbon at τ replaces each cost rate in* (5) *by the blended rate* $\text{rate}_\$ + \tau\,\text{rate}_{\text{kg}}$*, yielding the objective J of* (8)*. In the interior comparisons below, each holding the other decision fixed, the* direction *in which the carbon-minimizing choice departs from the cost-minimizing one is governed by four carbon-intensity-per-dollar ratios — holding η/h, inbound transport ε_0/t_1, last-mile ε_L/t_L, and facilities φ/F — while the magnitude and the realized joint optimum also depend on pooling, the bounds, and the complementarity of the two decisions.*

Proof. The blended rates in (8) follow by collecting coefficients in $K + \tau C$. The speed comparison sets holding against inbound transport at the margin (Proposition 2); the density comparison sets last-mile transport against a pooling-adjusted blend of facilities and holding (Proposition 4). The four ratios listed in the statement are therefore the primitive carbon-intensity comparisons used below. □

The carbon-minimizing choice on each decision therefore depends on each activity's carbon intensity *relative to its cost*, not on its absolute rates.

Writing $G(n, v) = \partial \text{SS}(n, v)/\partial n$, differentiation gives the stationarity conditions

$$F_\tau + H_\tau G(n, v) = \frac{\ell_\tau N \mu \delta}{2}\, n^{-3/2}, \tag{9}$$

$$N \mu D\, T'_\tau(v) = \frac{H_\tau\, \text{SS}(n, v)}{2v}, \tag{10}$$

the first balancing the marginal facility and lost-pooling costs against the marginal last-mile saving, and the second balancing the marginal transport cost of speed against the marginal holding saving it obtains. The joint first-order conditions thus reveal two marginal substitutions. The speed condition (10) governs the *availability margin*: safety stock and replenishment speed are alternative ways to provide service. The density condition (9) governs the *proximity margin*: facilities and last-mile distance are alternative ways to reach markets. The thresholds below compare the carbon intensity per dollar of the activities on each margin, and complementarity determines how the two margins interact in the joint configuration.

Availability: inventory or speed. The speed condition isolates the availability margin. Increasing speed raises inbound transport cost and carbon but lowers the safety stock required for service; the speed threshold identifies which activity is more carbon-intensive per dollar on this margin.

Proposition 2 (Speed reversal)**.** *Fix the density n and write the speed-dependent cost and carbon as*

$$\hat{K}(v) = h\,\kappa(n)\,v^{-1/2} + N\mu D\,t(v), \qquad \hat{C}(v) = \eta\,\kappa(n)\,v^{-1/2} + N\mu D\,\varepsilon(v),$$

with $\kappa(n) = z\sigma\sqrt{ND}\,\sqrt{\Psi(n)}$. If t and ε are increasing, differentiable, and convex, then at a common density and for interior optima the carbon-minimizing speed exceeds the cost-minimizing speed if and only if holding is more carbon-intensive per dollar than inbound transport at the margin*,*

$$\frac{\eta}{h} \;>\; \frac{\varepsilon'(v_{\mathrm{K}})}{t'(v_{\mathrm{K}})}, \tag{11}$$

evaluated at the cost-minimizing speed v_{K}. With the quadratic instantiation $\varepsilon(v) = \varepsilon_0 v^2$, $t(v) = t_0 + t_1 v^2$, the threshold reduces to $\eta/h > \varepsilon_0/t_1$ and the unconstrained optimum is closed form,

$$v^*(n) \;=\; \left[\frac{(h+\tau\eta)\,z\sigma\sqrt{ND}\,\sqrt{\Psi(n)}}{4\,(t_1+\tau\varepsilon_0)\,N\mu D}\right]^{2/5}. \tag{12}$$

On $[v_{\min}, v_{\max}]$ the optima are the projections of the unconstrained speeds onto the interval; under the threshold (11) *the comparison $v_C \geq v_K$ then holds weakly, and strictly when neither projection binds.*

Proof. $N\mu D\,t(v)$ and $N\mu D\,\varepsilon(v)$ are convex and $\kappa(n)v^{-1/2}$ has $\partial_v^2 = \frac{3}{4}\kappa(n)v^{-5/2} > 0$, so both objectives are strictly convex in v. The cost optimum satisfies $\hat{K}'(v_{\mathrm{K}}) = 0$, i.e. $N\mu D\,t'(v_{\mathrm{K}}) = \frac{1}{2}h\,\kappa(n)\,v_{\mathrm{K}}^{-3/2}$, so that $\frac{1}{2}\kappa(n)v_{\mathrm{K}}^{-3/2} = N\mu D\,t'(v_{\mathrm{K}})/h$. Evaluating the carbon derivative there,

$$\hat{C}'(v_{\mathrm{K}}) = -\eta\,\tfrac{1}{2}\kappa(n)v_{\mathrm{K}}^{-3/2} + N\mu D\,\varepsilon'(v_{\mathrm{K}}) = N\mu D\big(\varepsilon'(v_{\mathrm{K}}) - (\eta/h)\,t'(v_{\mathrm{K}})\big),$$

which is negative — so $v_C > v_{\mathrm{K}}$ by strict convexity — if and only if (11). With $t' = 2t_1 v$ and $\varepsilon' = 2\varepsilon_0 v$ the marginal ratio is ε_0/t_1, and the priced first-order condition (10) gives $v^{5/2} = (h+\tau\eta)z\sigma\sqrt{ND}\,\sqrt{\Psi}/[4(t_1+\tau\varepsilon_0)N\mu D]$, which is (12). The projection and the boundary qualification are immediate. □

Because (12) depends on the carbon price only through the blended ratio $(h+\tau\eta)/(t_1+\tau\varepsilon_0)$, which is monotone in τ, the fixed-density speed optimum moves monotonically between the cost and carbon endpoints as the carbon price rises: the endpoint comparison of the proposition also orders every finite price.

Total carbon is U-shaped in speed. For products whose inventory is emissions-intensive — refrigerated, volatile, or held to a high service level — a tighter carbon target can select *faster*

replenishment. The comparison is at common density; in the joint problem, the density channel reinforces the speed reversal when carbon also disperses the network and attenuates it otherwise (Proposition 5).

Service and volatility enter through the same channel. Write $J^{\mathrm{r}}(n;s) \equiv J\big(n, v^*(n;s); s\big)$, with $s = z\sigma$, for the reduced objective obtained after substituting the speed response into the priced objective.

Proposition 3 (Service and volatility as carbon parameters)**.** *Under the quadratic instantiation and for an interior speed response $v^*(n) \in (v_{\min}, v_{\max})$, $v^*(n)$ is increasing in the safety factor $z = \Phi^{-1}(\alpha)$ and in the demand standard deviation σ, and both raise the holding term in* (9)*. At an interior reduced optimum (reduced second-order condition $J^{\mathrm{r}}_{nn} > 0$), this is a local comparative-static result: a higher z or σ raises the inventory penalty of a denser network and shifts the carbon-minimizing density toward consolidation.*

Proof. Let $s = z\sigma$. From (12), $v^*(n;s) \propto s^{2/5}$, so speed rises with service and demand volatility. Along this speed response, $J^{\mathrm{r}}_n(n;s) = J_n\big(n, v^*(n;s); s\big)$ because $J_v = 0$. The holding marginal term in (9) is proportional to $s(v^*)^{-1/2}(1-\rho)/\sqrt{\Psi(n)} \propto s^{4/5}(1-\rho)/\sqrt{\Psi(n)}$, hence rises in s even after the induced speed increase. Thus $J^{\mathrm{r}}_{ns} > 0$. Under the reduced second-order condition $J^{\mathrm{r}}_{nn} > 0$, the implicit-function theorem gives $\mathrm{d}n^*/\mathrm{d}s = -J^{\mathrm{r}}_{ns}/J^{\mathrm{r}}_{nn} < 0$. □

A service requirement therefore affects the carbon-minimizing design as well as inventory cost; endogenous service choice is a natural extension (Section 7).

Proximity: facilities or last-mile travel. The density condition isolates the proximity margin. Increasing density shortens the last mile but raises facility cost and carbon and weakens pooling; the density threshold identifies when carbon places more weight on last-mile distance than on the facility-and-pooling cost of proximity, and pooling keeps the comparison from reducing to a single pair of rates.

Proposition 4 (Density and dispersion)**.** *Fix the speed v and let*

$$G(n,v) \;=\; \frac{\partial\,\mathrm{SS}(n,v)}{\partial n} \;=\; z\sigma\sqrt{D/v}\,\sqrt{N}\,\frac{1-\rho}{2\sqrt{\Psi(n)}} \;\geq\; 0 \tag{13}$$

be the marginal safety-stock penalty of an additional DC. The priced objective $J(\cdot, v)$ is strictly convex in n when

$$3\ell_\tau N\mu\delta\, n^{-5/2} \;>\; H_\tau z\sigma\sqrt{D/v}\sqrt{N}\,(1-\rho)^2\,\Psi(n)^{-3/2} \qquad \text{for all } n \in [1,N], \tag{14}$$

in which case the stationary point is unique; the minimizer is interior if and only if, in addition, $J_n(1) < 0 < J_n(N)$, and otherwise lies at a boundary, $n^* \in \{1, N\}$. When (14) *holds at $\tau = 0$ and in the limit $\tau \to \infty$, and for interior cost and carbon density optima, the carbon-minimizing*

density exceeds the cost-minimizing density, $n_C(v) > n_K(v)$, if and only if last-mile transport is more carbon-intensive per dollar than the pooling-adjusted facility threshold*,*

$$\frac{\varepsilon_L}{t_L} \;>\; \frac{\varphi + \eta\, G\big(n_K(v), v\big)}{F + h\, G\big(n_K(v), v\big)}. \tag{15}$$

The right-hand side is a weighted average of the facility ratio φ/F and the holding ratio η/h, with weight on η/h rising in the pooling penalty G: it equals φ/F when pooling is absent and tilts monotonically toward η/h as G grows (lower correlation, higher safety factor or volatility, or slower replenishment). In the no-pooling limit ($\rho \to 1$, so $G \to 0$) the optimum is closed form, $n^* = (\ell_\tau N\mu\delta/2F_\tau)^{2/3}$ (with the blended last-mile and facility rates $\ell_\tau = t_L + \tau\varepsilon_L$ and $F_\tau = F + \tau\varphi$ of (8)*), and* (15) *reduces to*

$$\frac{\varepsilon_L}{t_L} \;>\; \frac{\varphi}{F}. \tag{16}$$

The ordering $n_C(v) > n_K(v)$ is strict at interior optima; when an optimum lies at a boundary ($n = 1$ or N) it holds as a projected weak ordering.

Proof. Since

$$J_{nn} = \frac{3}{4}\,\ell_\tau N\mu\delta\, n^{-5/2} - \frac{1}{4}\,H_\tau z\sigma\sqrt{D/v}\sqrt{N}\,(1-\rho)^2\Psi(n)^{-3/2},$$

(14) gives strict convexity and the stated interior/boundary characterization. Because $(t_L + \tau\varepsilon_L)/(h + \tau\eta)$ is monotone in τ, the condition at the two endpoints implies it for every τ. For the threshold, take $\tau = 0$ for cost and $\tau \to \infty$ for carbon. The cost density satisfies $\partial_n K = F + h\,G(n_K, v) - t_L N\mu\delta/(2n_K^{3/2}) = 0$, so the last-mile marginal $t_L N\mu\delta/(2n_K^{3/2}) = F + h\,G(n_K, v)$. Evaluating the carbon derivative there,

$$\partial_n C\big|_{n_K} = \varphi + \eta\, G(n_K, v) - \frac{\varepsilon_L N\mu\delta}{2n_K^{3/2}} = \varphi + \eta\, G(n_K, v) - \frac{\varepsilon_L}{t_L}\big[F + h\,G(n_K, v)\big],$$

which is negative — so $n_C > n_K$ by convexity — if and only if (15). Writing $R(G) = (\varphi + \eta G)/(F + hG)$, we have $R(0) = \varphi/F$, $R(G) \to \eta/h$ as $G \to \infty$, and $R'(G) = (\eta F - h\varphi)/(F + hG)^2$, so R is monotone with $\operatorname{sign} R' = \operatorname{sign}(\eta/h - \varphi/F)$ and lies between φ/F and η/h. In the no-pooling limit $\rho \to 1$, $G \to 0$ and $\Psi \to N$, so (9) gives $n^* = (\ell_\tau N\mu\delta/2F_\tau)^{2/3}$ and (15) becomes (16). □

The density decision is the centralize-versus-decentralize question in carbon terms. By (15), the carbon-minimizing network is more dispersed when the last-mile ratio ε_L/t_L exceeds a facility threshold that pooling shifts from φ/F toward η/h. Pooling therefore enters as an explicit weight on the holding ratio, not as a uniform centralizing force: when holding is more carbon-intensive per dollar than facilities, stronger pooling raises the threshold for dispersion; otherwise it lowers it. A corner optimum, when it arises, is fully centralized or fully decentralized.

Complementarity of speed and density. The two margins interact through safety stock: faster replenishment reduces the inventory penalty of density, and higher density increases the

inventory exposure affected by speed. Proposition 5 makes the interaction precise.

Proposition 5 (Complementarity)**.** *$v^*(n)$ is increasing in n, and under the density second-order condition $n^*(v)$ is increasing in v.*

Proof. Since $G(n, v) \propto v^{-1/2}$, the cross-partial is $J_{nv} = H_\tau G_v = -H_\tau G(n, v)/(2v) < 0$ for $\rho < 1$. For the speed response, J is strictly convex in v (as in the proof of Proposition 2), so by the implicit function theorem $\mathrm{d}v^*/\mathrm{d}n = -J_{nv}/J_{vv} > 0$. For the density response, under the second-order condition $J_{nn} > 0$ of (14), $\mathrm{d}n^*/\mathrm{d}v = -J_{nv}/J_{nn} > 0$. □

Faster replenishment shrinks safety stock and lowers the inventory penalty of dispersion; a denser network raises the inventory at stake and raises the value of speed. Speed investments and network density are therefore best evaluated jointly.

The joint regime. The two marginal comparisons can point in any combination of directions. The corollary below records a sufficient condition under which the carbon best-response equilibrium lies weakly above the cost optimum in both network density and replenishment speed.

Corollary 6 (Faster and more dispersed)**.** *Suppose, over the relevant range of configurations, the speed threshold of Proposition 2 holds at every density and the pooling-adjusted density threshold* (15) *of Proposition 4 holds at every speed, so that the carbon-minimizing best responses dominate the cost-minimizing ones pointwise, $v_C(n) \geq v_K(n)$ for all n and $n_C(v) \geq n_K(v)$ for all v. Suppose also that the density second-order condition* (14) *holds at the carbon rates and that the carbon best responses are single-valued and continuous on the interval reached by the iteration. Then best-response iteration from the cost optimum converges to a carbon best-response equilibrium that is weakly faster and weakly more dispersed than the cost-minimizing configuration, strictly so at interior optima.*

Proof. The two thresholds give the pointwise dominance $v_C(n) \geq v_K(n)$ and $n_C(v) \geq n_K(v)$ (Propositions 2 and 4), and by Proposition 5 each best response is nondecreasing ($v^*(\cdot)$ in n, $n^*(\cdot)$ in v). Starting from the cost optimum (n_K, v_K), a fixed point of the cost best responses, and iterating the carbon best responses gives a nondecreasing sequence: $v_C(n_K) \geq v_K(n_K) = v_K$, then $n_C\big(v_C(n_K)\big) \geq n_C(v_K) \geq n_K(v_K) = n_K$, and so on. The iterates are monotone and bounded, hence converge; by continuity of the single-valued best-response maps, the limit is a fixed point of the carbon best-response map and is weakly no smaller than (n_K, v_K) (Topkis 1998), with strict inequalities at interior optima. □

The corollary is a best-response result under the stated conditions. A global optimum statement would require convexity over the whole feasible box $[1, N] \times [v_{\min}, v_{\max}]$; the speed–pooling cross term $J_{nv} = H_\tau G_v$ can exceed the own curvatures in the extreme high-density, low-speed corner.
In the calibrated illustrations, the finite-grid computation finds the stated configuration as the lowest sampled priced objective and verifies positive Hessian minors at the sampled points around

the reported optimum. These checks document the sampled configurations used in Section 5 rather than extend the analytical corollary.

A refrigerated, volatile, high-service product delivered by diesel van illustrates this regime: its carbon-minimizing chain replenishes faster and holds stock closer to demand. The contribution is the diagnostic structure, not the universality of this regime (Section 6).

Demand uncertainty: a chance-constrained cap. The cost and carbon functions are expected per-period flows, so the cap in (7) bounds *expected* carbon. Because demand is random, realized carbon fluctuates about $C(n, v)$: the facility charge and the base-stock safety-stock target are planned and treated as deterministic, but inbound and last-mile freight scale with realized throughput $Q = \sum_i X_i$, the sum of the realized market demands, which has mean $N\mu$ and variance $\sigma_Q^2 = \sigma^2 N\,[\,1 + (N-1)\rho\,]$. Realized carbon is then

$$\tilde{C}(n, v) \;=\; \varphi n + \eta\,\mathrm{SS}(n, v) + \big[\varepsilon(v)D + \varepsilon_L d(n)\big]\,Q, \tag{17}$$

and a firm that requires realized carbon to stay within B with confidence $1 - \beta_c$ — where β_c is the carbon-side tail probability, written with the subscript c to distinguish it from the inventory fill-rate β of the lot-sizing literature — imposes the chance constraint $\Pr\{\tilde{C}(n, v) \le B\} \ge 1 - \beta_c$, the carbon-side counterpart of the cycle service level α on the inventory side.

Proposition 7 (Deterministic equivalent of the chance constraint)**.** *This chance constraint is equivalent to*

$$\varphi n + \eta\,\mathrm{SS}(n, v) + m\big[\varepsilon(v)N\mu D + \varepsilon_L N\mu\, d(n)\big] \;\le\; B, \tag{18}$$

the carbon function (6) *with the transport component scaled by* $m = q_{1-\beta_c}/(N\mu)$*, where* $q_{1-\beta_c}$ *is the* $(1-\beta_c)$ *quantile of realized throughput* Q*, and the facility and holding terms unchanged. Under the normal approximation* $Q \sim \mathcal{N}(N\mu, \sigma_Q^2)$ *this is* $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$*, with* $z_{\beta_c} = \Phi^{-1}(1-\beta_c)$ *and* $\mathrm{CV}_Q = \sigma_Q/(N\mu)$*; then* $m \ge 1$ *whenever* $\beta_c \le \frac{1}{2}$*.*

Proof. In (17) the throughput coefficient $\gamma(n, v) = \varepsilon(v)D + \varepsilon_L d(n) > 0$, so $\{\tilde{C} \le B\} = \{Q \le (B - \varphi n - \eta\mathrm{SS})/\gamma\}$, and $\Pr\{\tilde{C} \le B\} \ge 1 - \beta_c$ holds iff $(B - \varphi n - \eta\mathrm{SS})/\gamma \ge q_{1-\beta_c}$, i.e. $\varphi n + \eta\mathrm{SS} + \gamma\, q_{1-\beta_c} \le B$. Writing $\gamma\, q_{1-\beta_c} = (q_{1-\beta_c}/N\mu)\,[\varepsilon(v)N\mu D + \varepsilon_L N\mu d(n)]$ gives (18). Under normality $q_{1-\beta_c} = N\mu + z_{\beta_c}\sigma_Q$, so $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$. □

The chance constraint therefore leaves the structure intact: (18) is the four-component carbon function with the inbound and last-mile carbon rates scaled, $\varepsilon_0 \mapsto m\varepsilon_0$ and $\varepsilon_L \mapsto m\varepsilon_L$, while cost and the facility and holding terms are unchanged. Because Q is realized market throughput, exogenous to the firm's choice of n and v, the factor m is a decision-invariant scalar on the transport carbon rates; the priced objective is again (8) under this substitution, so Propositions 2–5 and Corollary 6 carry over, and only the two thresholds shift.

Corollary 8 (Uncertainty tightens the speed threshold and relaxes the density threshold)**.** *Under the chance-constrained cap the speed threshold* (11) *becomes* $\eta/h > m\,\varepsilon'(v_{\mathrm{K}})/t'(v_{\mathrm{K}})$ *(*$m\,\varepsilon_0/t_1$ *under*

the quadratic), and the pooling-adjusted density threshold (15) *becomes*

$$m\,\frac{\varepsilon_L}{t_L} \;>\; \frac{\varphi+\eta\,G\big(n_K(v),v\big)}{F+h\,G\big(n_K(v),v\big)}, \tag{19}$$

which reduces to $m\,\varepsilon_L/t_L > \varphi/F$ *in the no-pooling limit* ($\rho \to 1$)*. Under the normal approximation,* $m \geq 1$ *for* $\beta_c \leq \frac{1}{2}$*, with equality at* $\beta_c = \frac{1}{2}$*, so the chance constraint tightens the speed-reversal threshold and relaxes the dispersion threshold relative to the expected-value cap, and the effect grows as* m *rises with the demand coefficient of variation* σ/μ*, the correlation* ρ*, and the confidence* $1-\beta_c$*.*

Proof. The chance constraint multiplies the inbound and last-mile carbon rates by m (Proposition 7); substituting $m\varepsilon_0$ and $m\varepsilon_L$ for ε_0 and ε_L in (11) and (15) gives the two conditions, the latter scaling only the left side and leaving the pooling-adjusted facility threshold $(\varphi+\eta G)/(F+hG)$ on the right. From $\sigma_Q^2 = \sigma^2 N[1+(N-1)\rho]$, $\mathrm{CV}_Q = (\sigma/\mu)\sqrt{[1+(N-1)\rho]/N}$ is increasing in σ/μ and, for $N > 1$, in ρ, and z_{β_c} is increasing in $1-\beta_c$, so $m = 1+z_{\beta_c}\mathrm{CV}_Q$ is increasing in all three. □

Demand uncertainty therefore affects carbon through two channels. It enters the inventory-holding component through the safety stock $z\sigma\sqrt{L}$, the channel behind the speed reversal, and, under a chance-constrained cap, it enters the transport component through the margin $z_{\beta_c}\,\mathrm{CV}_Q$ that the firm holds against realized emissions just as it holds safety stock against demand. More variable or more correlated demand raises this margin and, like a higher service requirement, shifts the carbon-minimizing configuration; the expected-value cap is the limiting case $m = 1$. Carrying the realized variability of on-hand inventory as well would place an analogous margin on the inventory-holding component; we keep the safety stock at its planned level, consistent with the base-stock policy. The deterministic equivalent itself is distribution-free in the throughput quantile $q_{1-\beta_c}$; the normal approximation enters only through the closed form $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$, which is most accurate for moderate N and modest CV_Q and which a firm with strongly skewed demand can replace by the empirical throughput quantile without altering the structure.

The cap applies over the same accounting period as the flow model — here a week. For a cap on cumulative carbon over T periods with demands independent across periods, aggregate throughput has coefficient of variation $\mathrm{CV}_Q/\sqrt{T}$, so the multiplier tightens toward one as the horizon lengthens: at the baseline calibration, the 95% multiplier falls from $m = 1.37$ for a weekly cap to $m = 1.05$ for an annual cap ($T = 52$). The chance-constrained results are therefore most relevant to high-frequency operational carbon budgets, shipment-cycle accounting, and corporate carbon controls, and less binding for annual compliance accounting.

In-transit inventory. The holding term is modeled through safety stock; in-transit (pipeline) inventory, $N\mu L = N\mu D/v$, falls in v and is omitted from our model. If pipeline inventory were carried at the warehouse holding rates, it would add $(h+\tau\eta)\,N\mu D/v$ to the holding term; the speed first-order condition then acquires a v^{-2} term, and stationarity again equates $(h+\tau\eta)/(t_1+\tau\varepsilon_0)$ to a quantity strictly increasing in v — the same ratio that governs (11). Valued at common

rates, pipeline inventory therefore shifts the level of v^* (the closed form (12) is replaced by a numerical root) but preserves the carbon-intensity threshold; carrying capital cost with negligible warehouse carbon instead raises only the cost-minimizing speed, shifting the threshold against the speed reversal. We treat pipeline inventory as a level effect under the equal-rate convention and leave alternative carbon accounting for in-transit stock as an extension.

Load-factor sensitivity. Our model treats the per-unit last-mile rates t_L and ε_L as independent of shipment size. Last-mile cost and carbon, however, accrue largely per *vehicle*-mile — driver time and fuel — so the per-unit rates scale with vehicle-miles per unit, which rise as dispersion thins each territory's dispatch volume and average vehicle fill falls; load-sensitive rates therefore introduce a consolidation force outside the constant-rate model. Because a fill multiplier depends on n, it enters the marginal density condition through both the level and the derivative of the last-mile term. When a common fill multiplier is applied to the cost and the carbon rate alike, both optima shift toward consolidation in the reported computation, while their relative direction remains consistent with the carbon-intensity comparison. Per vehicle-mile, cost is dominated by driver time while fuel use falls somewhat in an emptier vehicle — the basis of tonne-kilometre emission accounting (Smart Freight Centre 2022) — so when fill deteriorates, the per-unit cost rate typically inflates at least as strongly as the per-unit carbon rate. A multiplier falling more heavily on carbon is therefore a stress case for the dispersion result; under it, the density comparison can shift toward consolidation. Section 5 reports a reduced-form sensitivity computation for both cases; a full treatment would make shipment size, routing, and consolidation additional decisions, which we leave as an extension.

5 Numerical Illustrations

This section applies the analytical thresholds to calibrated products, reports the implied designs and cost–carbon trade-offs, and shows how the mechanisms interact under the maintained model assumptions. The dollar magnitudes are chosen so that the cost-minimizing network is interior, and the four carbon-intensity-per-dollar ratios are set to regime-representative values anchored in published logistics ranges — line-haul and last-mile transport, refrigerated and ambient storage, and facility energy. All reported quantities are computed from the cost and carbon functions of Sections 3–4 by a finite-grid computation over the two decision variables n and v.

5.1 Calibration

Table 2 lists the baseline parameters. Units are normalized: the period is a week, and the inbound leg is the unit of distance ($D = 1$), so t_0, t_1, and ε_0 are per-unit coefficients over that leg rather than per mile, while the last-mile rates t_L and ε_L are per unit-mile against the $d(n) = \delta n^{-1/2}$ miles of the delivery leg. The four ratios give $R_{\text{speed}} = (\eta/h)/(\varepsilon_0/t_1) = 0.375$ and $R_{\text{density}} = (\varepsilon_L/t_L)/(\varphi/F) = 0.75$, placing the baseline in the slower-and-more-consolidated regime, where

Table 2: Baseline calibration. Dollar magnitudes place the cost optimum in the interior; the four carbon-intensity-per-dollar ratios are set to regime-representative values anchored in published logistics ranges.

Group	**Values**
Demand, geography	$N = 40$, $\mu = 20{,}000$, $\sigma = 8{,}000$ (CV 0.40), $\rho = 0.30$, $\alpha = 0.95$ ($z = 1.645$), $\delta = 120$, $D = 1$, $v \in [0.3, 3]$
Cost rates ($)	$F = 7500$, $h = 0.50$, $t_0 = 1.00$, $t_1 = 0.05$, $t_L = 0.005$
Carbon rates (kg)	$\varphi = 3000$, $\eta = 0.15$, $\varepsilon_0 = 0.04$, $\varepsilon_L = 0.0015$
Ratios	$\eta/h = 0.30$, $\varepsilon_0/t_1 = 0.80$, $\varepsilon_L/t_L = 0.30$, $\varphi/F = 0.40$
Diagnostics	$R_{\text{speed}} = 0.375$, $R_{\text{density}} = 0.75$

both comparisons fall short of one. Solving (7), the cost-minimizing configuration is $n^* = 7.8$ distribution centers replenished at speed $v^* = 1.03$, with weekly cost \$1.24M and carbon 160.5 t, divided between a transport component of 85.6 t and a facility-and-holding component of 74.9 t. The carbon-minimizing configuration is slower and more consolidated, $n^* = 6.6$ and $v^* = 0.69$, lowering carbon by 5.1% at a cost increase of 1.2%. The movement is modest because the two components substitute: slowing or consolidating shifts carbon from one component to the other rather than removing it, so for a product in this regime the net change is small. The magnitude of available abatement is regime-dependent, as the archetypes below show. The strict-convexity condition (14) compares last-mile curvature with the holding pooling-loss term, and is favored by a geographically spread region (large δ) or lower demand variability (small σ/μ); it is weakest when pooling is strong and the region compact.[2] It holds at every configuration reported in this section — the baseline and all four archetypes — so each density optimum is the unique interior stationary point.

5.2 The four regimes

Holding the cost rates fixed and sweeping the two carbon-intensity ratios traces the analytical thresholds. Figure 2 plots the computed carbon-minimizing direction in the $(R_{\text{density}}, R_{\text{speed}})$ plane: the solid curves are the speed and density flip boundaries at $\rho = 0.30$, the dashed lines the no-pooling thresholds $R_{\text{speed}} = 1$ and $R_{\text{density}} = 1$. Without pooling the two boundaries separate exactly at $R = 1$, splitting the plane into the four regimes (set out, with their managerial readings, in Table 6 of Section 6). With pooling the speed boundary stays near $R_{\text{speed}} = 1$, while the density boundary tilts further right as R_{speed} rises: by the pooling-adjusted threshold (15), dispersion requires the last-mile ratio to clear a facility threshold that pooling pulls toward η/h, so the boundary lies to the right of $R_{\text{density}} = 1$ where holding is more carbon-intensive per dollar than facilities ($\eta/h > \varphi/F$) and near or left of it otherwise. The computed boundary thus traces the mediant — the ratio

[2] Explicitly, at a given speed v the binding case of (14) is $n = N$, where it reduces to $\delta > H_\tau z\sigma\sqrt{D/v}\sqrt{N}(1-\rho)^2/[3\ell_\tau\mu]$. At the baseline this requires $\delta > 82$ at the carbon optimum and $\delta > 67$ at the cost optimum, both met by $\delta = 120$; only the slowest speeds at the highest density (the non-convex corner noted after Corollary 6) require a larger region.

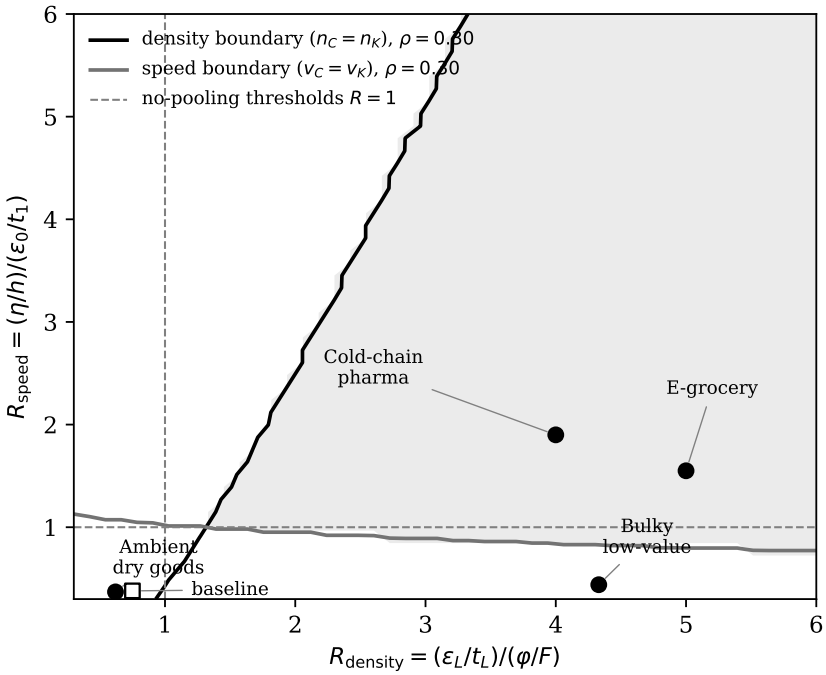


Figure 2: Computed regime boundaries in the plane of the two carbon-intensity-per-dollar ratios. Solid curves are the speed boundary ($v_C = v_K$) and the density boundary ($n_C = n_K$) at $\rho = 0.30$; dashed lines are the no-pooling thresholds $R_{\text{speed}} = 1$ and $R_{\text{density}} = 1$. The pooling-adjusted density boundary tilts further right as R_{speed} rises, lying to the right of $R_{\text{density}} = 1$ where $\eta/h > \varphi/F$ and near or left of it otherwise, tracing the mediant of (15). The shaded region is the faster-and-more-dispersed regime; points locate the four archetypes of Table 3 and the baseline.

of summed numerators to summed denominators — of the facility and holding comparisons in Proposition 4.

5.3 Product archetypes

The four archetypes are guided by external benchmarks: the refrigerated-warehouse energy premium places cold-chain and e-grocery goods in the high-η/h band; low-fill diesel vans raise ε_L/t_L for e-grocery, pharmaceutical, and bulky goods; value density enters through the capital component of h; and the line-haul mode factors order inbound carbon against speed.[3] The emission factors are well established, while the dollar denominators are normalized order-of-magnitude cost proxies, so

[3]Refrigerated warehouses run at roughly 3.7× the ambient energy intensity (≈ 8.6 versus 2.3 kg $CO_2e/ft^2/yr$) (U.S. Environmental Protection Agency 2024, 2025). A delivery van emits ≈ 0.62 kg CO_2e/t-km at 37–41% fill, roughly 6–8× an articulated HGV (DESNZ/DEFRA 2024b,a; Smart Freight Centre 2022). Value density spans crushed stone at ≈ \$17.5/tonne (U.S. Geological Survey 2025) to pharmaceuticals at ≈ \$49,000/tonne (U.S. Federal Highway Administration and Bureau of Transportation Statistics 2023); because the capital cost of inventory is part of h, cold-chain pharma's high η/h assumes its refrigeration and compliance energy dominates the capital denominator. Line-haul factors order ocean 0.016 < rail 0.028 < road 0.10–0.21 kg CO_2e/t-km (DESNZ/DEFRA 2025); ε_0/t_1 is not separately identified. Supporting value, fill, and cost benchmarks are from World Bank (2023); U.S. Census Bureau and Bureau of Transportation Statistics (2015); Cushman & Wakefield (2026). The assumed ratios (η/h, ε_0/t_1, ε_L/t_L, φ/F) are $(0.30, 0.80, 0.25, 0.40)$ for ambient dry goods, $(0.95, 0.50, 1.40, 0.35)$ for cold-chain pharma, $(0.20, 0.45, 1.30, 0.30)$ for bulky low-value goods, and $(0.85, 0.55, 1.50, 0.30)$ for e-grocery perishables.

Table 3: The four product archetypes: the diagnostics $R_{\text{speed}} = (\eta/h)/(\varepsilon_0/t_1)$ and $R_{\text{density}} = (\varepsilon_L/t_L)/(\varphi/F)$, the carbon-minimizing configuration (n_C, v_C) and its direction relative to the shared cost optimum ($n_K = 7.8$, $v_K = 1.03$), the largest attainable carbon reduction, and the cost premium at the carbon minimum.

Archetype	R_{speed}	R_{density}	Carbon-min direction	(n_C, v_C)	Max. reduction	Cost @ min
Ambient dry goods	0.375	0.625	slower, consolidate	(5.8, 0.69)	6.1%	+1.5%
Cold-chain pharma	1.90	4.00	faster, disperse	(16.5, 1.42)	7.7%	+3.8%
Bulky low-value	0.44	4.33	slower, disperse	(22.8, 0.82)	17.4%	+7.8%
E-grocery perishable	1.55	5.00	faster, disperse	(18.9, 1.32)	9.6%	+4.8%

the benchmarks set the *relative* magnitudes of the ratios across archetypes; the chosen values are regime-representative calibrations that span the four regimes.

Table 3 reports the diagnostics and the carbon-minimizing outcomes. Because the archetypes share the cost rates, all four have the same cost-minimizing configuration ($n^* = 7.8$, $v^* = 1.03$) and differ only in the direction and magnitude by which the carbon-minimizing configuration departs from it. Slowing and consolidating is carbon-minimizing for the ambient dry-goods archetype; the cold-chain and e-grocery archetypes call for faster and more dispersed networks; and the bulky, low-value archetype, with inexpensive storage but a carbon-intensive last mile, disperses while slowing. Available abatement ranges from about 6% to 17% of total carbon across the four; reaching 80% of that abatement costs between 0.5% and 2.2%, while reaching the carbon minimum costs between 1.5% and 7.8%, the larger premiums attaching to the products that disperse most. All four archetypes have interior density optima ($J_n(1) < 0 < J_n(N)$), so none is a corner solution.

5.4 Value of joint redesign

The two decisions are complements (Proposition 5); in this illustration, adjusting them together reaches abatement levels and target costs that adjusting either alone does not reach. Table 4 reports this for a product in the faster-and-more-dispersed regime (e-grocery perishables, $R_{\text{speed}} = 1.55$, $R_{\text{density}} = 5.0$). Holding the network fixed and varying only speed lowers carbon by at most 0.5%; holding speed fixed and varying only density reaches 8.4%; adjusting both reaches 9.6%. We report two targets: a *moderate* target of 7.6%, set at 90% of the largest abatement attainable by adjusting one decision alone (here density, $0.90 \times 8.4\%$), and a *deep* target of 9.1%, set at 95% of the joint attainable abatement ($0.95 \times 9.6\%$). The moderate target is infeasible by adjusting speed alone and costs 1.8% by adjusting density alone, but only 1.3% when both are adjusted; the deep target lies beyond either single adjustment and is met only by adjusting both. In this illustration, the speed margin has limited isolated abatement value, but it expands the cost-effective range of density adjustment by reducing the inventory penalty of dispersion.

Table 4: Value of adjusting both decisions, for the e-grocery archetype. "Max. reduction" is the largest carbon reduction the policy attains; the last two columns give the cost increase to meet a moderate and a deep carbon target.

Policy	Max. reduction	Cost at 7.6% target	Cost at 9.1% target
Speed only (n fixed)	0.5%	infeasible	infeasible
Density only (v fixed)	8.4%	+1.8%	infeasible
Joint (n, v)	9.6%	+1.3%	+2.8%

5.5 Chance-constrained carbon caps

Moving from an expected carbon cap to a cap on *realized* carbon held with high confidence is not merely a tightening of the same constraint; by Proposition 7 it scales the transport component by $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$ and so reweights transport emissions relative to inventory and facilities, changing the relative attractiveness of speed and density. We trace this for the baseline product. With $N = 40$, $\mu = 20{,}000$, $\sigma = 8{,}000$ (CV 0.40), and $\rho = 0.30$, the aggregate throughput coefficient of variation is

$$\mathrm{CV}_Q = \frac{\sigma}{\mu}\sqrt{\frac{1 + (N-1)\rho}{N}} \approx 0.225,$$

so the transport scalar is $m \approx 1.29$, 1.37, and 1.52 at the 90%, 95%, and 99% confidence levels ($z_{\beta_c} \approx 1.28$, 1.64, 2.33), against $m = 1$ for the expected cap.

Table 5 reports the carbon-minimizing design under the scaled carbon function at each confidence level, with ε_0 and ε_L replaced by $m\varepsilon_0$ and $m\varepsilon_L$. Two effects appear, consistent with Corollary 8. On speed, the effective inbound ratio $m\varepsilon_0/t_1$ rises from 0.80 to 1.22, moving further above the holding ratio $\eta/h = 0.30$, so the carbon-minimizing speed falls at every confidence level, from 0.69 to 0.59. On density, the effective last-mile ratio $m\varepsilon_L/t_L$ rises from 0.30 to 0.46, raising the risk-weighted last-mile term against the pooling-adjusted facility threshold of Proposition 4. That proposition compares density best responses at a *fixed* speed; the table instead reports the *joint* carbon-minimizing density n_C against the joint cost-minimizing density n_K, in which the speed also adjusts. By the fixed-speed test, dispersion would set in already at the 90% cap, where $m\varepsilon_L/t_L = 0.39$ clears the pooling-adjusted threshold (≈ 0.37 at the cost optimum); on the joint basis the risk margin also lowers the speed, and by complementarity (Proposition 5) a slower chain calls for fewer warehouses, which partly offsets the dispersion, so the joint n_C rises through n_K later, between the 90% and 95% levels. For this product a high-confidence carbon cap therefore calls for a chain that is slower and progressively less consolidated, and beyond a threshold confidence more dispersed than the cost optimum — a change in direction produced by the accounting rule alone, with the physical parameters held fixed. Demand uncertainty thus enters the carbon design through two channels: safety stock on the inventory side and a throughput-risk margin on the transport side. Figure 3 traces the two designs across confidence levels.

More variable demand deepens these effects: repeating the 95% cap at demand CVs of 0.20, 0.40, and 0.60 raises m from 1.19 to 1.56, widens the speed reduction, and shifts the network from

Table 5: Effect of chance-constrained carbon accounting on the carbon-minimizing design (baseline product; cost optimum $n_K = 7.8$, $v_K = 1.03$; reference thresholds $\eta/h = 0.30$ and the no-pooling facility ratio $\varphi/F = 0.40$, against which the pooling-adjusted density threshold at the cost optimum is ≈ 0.37). "Speed" and "Network" give the direction relative to the cost optimum. Higher confidence scales the transport component by m, lowering the carbon-minimizing speed and, beyond the 90% level, dispersing the network.

Cap / conf.	m	$m\varepsilon_0/t_1$	$m\varepsilon_L/t_L$	v_C	n_C	Speed	Network	Cost prem.
Expected (m=1)	1.00	0.80	0.30	0.69	6.6	slower	consolidate	+1.2%
90%	1.29	1.03	0.39	0.63	7.7	slower	consolidate	+1.7%
95%	1.37	1.10	0.41	0.62	8.0	slower	disperse	+1.9%
99%	1.52	1.22	0.46	0.59	8.6	slower	disperse	+2.2%

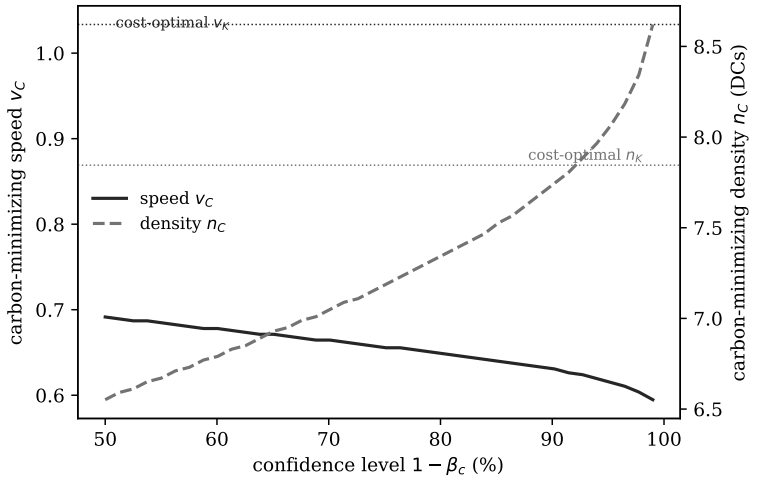


Figure 3: Carbon-minimizing speed v_C (left axis, solid) and density n_C (right axis, dashed) under increasingly stringent confidence levels for the baseline product. Dotted lines mark the cost-optimal speed v_K and density n_K; the carbon-minimizing speed stays below v_K throughout, while the carbon-minimizing density rises through n_K between the 90% and 95% confidence levels.

relatively consolidated to relatively dispersed as the cost optimum consolidates with variability.[4]

5.6 Robustness

The robustness analysis comprises four checks. Restricting the network to an integer number of DCs returns the continuous result rounded: cost (8, 1.04), carbon (7, 0.69). Restricting replenishment to three discrete modes ($v \in \{0.5, 1.0, 2.0\}$) leaves the cost optimum at the standard mode, (7.8, 1.0), and shifts the carbon optimum to the slow mode, (6.3, 0.5), the discrete counterpart of the continuous speed reduction. As the demand correlation rises from $\rho = 0$ to $\rho = 0.9$, both optima add facilities as pooling weakens — the cost-minimizing density from 5.9 to 9.8 and the carbon-minimizing density from 4.8 to 8.1 — with the carbon optimum consolidated relative to the

[4] A Monte Carlo check with 5×10^5 throughput draws returns the nominal violation rates under normally distributed throughput and shows under-coverage under right-skewed throughput; in such cases the firm can set z_{β_c} from the empirical throughput quantile, as the distribution-free form of Proposition 7 permits.

cost optimum throughout. The fourth check is the load-factor sensitivity of Section 4, computed with the reduced-form fill multiplier $0.5 + 0.5\,(n/n_{\text{ref}})$, $n_{\text{ref}} = 8$ (half load-sensitive at the reference density). Applying the multiplier to the carbon rate only — the stress case of Section 4 — lowers the carbon-minimizing density sharply, from 6.6 to 3.0 DCs for the baseline and from 22.8 to 5.8 for the bulky archetype, reversing the bulky archetype's dispersion relative to the unadjusted cost optimum. Applying the same multiplier to the cost and carbon rates alike consolidates the cost optimum as well, from 7.8 to 3.4 DCs, and in this computation each product's direction is preserved: the baseline still consolidates and the bulky archetype still disperses relative to the load-adjusted cost optimum. Load sensitivity thus shifts both optima toward consolidation; in these illustrations, reversal occurs only in the stress case, when the fill multiplier falls more heavily on carbon.

6 Managerial Implications

Table 6 translates the two marginal comparisons into a product-level diagnostic map. The rows describe the availability margin (holding against inbound transport); the columns describe the proximity margin (last-mile delivery against facilities). Each cell indicates the direction of carbon pressure relative to the cost optimum; slowing and consolidating correspond to the lower-left cell. The final configuration follows from the joint speed–density design problem with these marginal forces in view.

Using the diagnostic ratios. A firm can use the ratios, estimated from its own rate data, as diagnostics for the joint design problem. The holding-to-inbound comparison indicates the carbon pressure on the availability margin (Proposition 2). The last-mile-to-facility comparison, adjusted for pooling, indicates the carbon pressure on the proximity margin: pooling shifts the threshold from φ/F toward the holding ratio η/h, raising the bar for dispersing when $\eta/h > \varphi/F$ (Proposition 4). Because speed changes the inventory cost of density, and density changes the inventory value of speed, the two margins are evaluated together in the final configuration (Proposition 5). Under a cap on realized carbon, the transport intensities are first scaled by the throughput-risk multiplier (Proposition 7).

Replenishment speed. For refrigerated, volatile, or high-service products, a tighter carbon target can be met through faster shipping: the safety stock a slower mode would require carries more carbon than the faster transport adds (the top row of Table 6).

Consolidation and the last mile. Consolidation lowers carbon here through inventory pooling and reduced facility energy while lengthening the last mile. A firm whose last mile is carbon-intensive — diesel vans, long rural routes — can lower total carbon by holding more stock nearer to demand, even though that raises total inventory; the transport channel of consolidation works against it. This last channel is strong when vehicle fill improves with consolidation, and the load-factor sensitivity of Section 4 separates its two effects: a fill multiplier common to cost and

Table 6: Diagnostic map of carbon pressure relative to the cost optimum: rows give the availability margin, columns the proximity margin. The column split is exact in the no-pooling limit and is otherwise the pooling-adjusted threshold of Proposition 4.

	Lower network density *last mile less carbon-intensive per dollar than facilities*	**Higher network density** *last mile more carbon-intensive per dollar than facilities*
Higher replenishment speed *holding more carbon-intensive per dollar than inbound transport*	faster and more consolidated, pooling inventory upstream — carbon-intensive inventory with a lower-emission or short last mile	*faster and more dispersed* — cold-chain, volatile, high-service goods on a diesel last mile
Lower replenishment speed *holding less carbon-intensive per dollar than inbound transport*	slower and more consolidated — the transport-side case; ambient, stable, low-service dry goods	slower long haul, denser network — lower-emission, low-cost storage with carbon-intensive inbound and last mile

carbon consolidates both optima together, while a multiplier that falls more heavily on carbon — a stress case, since per-vehicle-mile cost is typically at least as fill-sensitive as fuel — can shift the density comparison toward consolidation — in the carbon-only computation, the bulky archetype's carbon-minimizing network falls from 22.8 to 5.8 DCs (Section 5). A firm should therefore weigh its last mile's fill economics — and whether fill degrades its emissions faster than its cost — before dispersing.

Joint adjustment of the two margins. Because the margins are complements, a program that acts on only one margin — a slow-steaming mandate or a consolidation initiative — attains part of the available abatement, and for some products presses a margin against its carbon-minimizing direction: for the e-grocery archetype, speed alone abates at most 0.5% and density alone 8.4%, while the joint design reaches 9.6% and meets a common target at lower cost (Table 4).

Product segmentation, service, and volatility. Because the carbon-minimizing configuration depends on the product, a single chain-wide policy serves a heterogeneous portfolio imperfectly. The service-level target and demand variability also bear on carbon: a higher service-level target and a more volatile demand both enlarge safety stock and so, by Proposition 3, raise the carbon-minimizing speed and strengthen the pull toward consolidation through the value of pooling, unless the last-mile comparison prevails in the density decision. A 99% cycle-service target is thus a carbon-design parameter, not only an inventory-cost one. Higher demand correlation across markets weakens the pooling benefit that favors consolidation and can make dispersion more attractive, so forecasting and demand-smoothing bear on carbon in their own right.

Carbon accounting and confidence. Whether the cap binds expected or realized carbon is itself a design parameter: a high-confidence cap places a risk margin on transport emissions, tightening the speed threshold and relaxing the dispersion threshold (Corollary 8), and for the baseline product it shifts the network from consolidation to dispersion beyond a threshold confidence (Table 5).

Implication for policy. Our results also suggest a caution for regulation. A rule that prices or caps only transport leaves the facility-and-holding component untouched and can induce substitution into inventory and facilities, potentially raising emissions the rule does not measure. Formally, a transport-only instrument prices $C_{\mathrm{T}}(n, v) = \varepsilon(v)\, N\mu D + \varepsilon_L\, N\mu\, d(n)$; minimizing $K + \tau C_{\mathrm{T}}$ blends the carbon price into the transport rates alone, so the induced response can raise the unpriced holding and facility carbon. In the baseline calibration a transport-only price lowers transport carbon while raising total carbon. Instruments scoped to transport alone — a freight fuel levy, a mode-shift mandate, or a vehicle-emissions standard — are exposed to this substitution. An instrument that prices a firm's total logistics carbon (for example, a scope-inclusive carbon charge or a cap covering both warehousing and transport emissions) prices the transport, facility, and inventory components and lets the firm choose the speed-and-density mix that minimizes carbon across them.

7 Discussion and Conclusion

We have studied a distribution network that provides two services, availability and proximity, each of which the firm can provide through more than one operational activity: safety stock or replenishment speed for availability, facilities or last-mile distance for proximity. The carbon-minimizing configuration depends on which activity provides each service at lower carbon intensity per dollar. The contribution of the paper is to make these two substitutions explicit and to give the thresholds at which the carbon-minimizing configuration departs from the cost-minimizing one.

Limitations. The thresholds apply to an average, symmetric distribution environment, so their use in a heterogeneous network is diagnostic rather than mechanical. Heterogeneous markets or discrete facility sites replace $d(n) = \delta n^{-1/2}$ with the realized last-mile profile; this can move the density threshold, but it leaves the underlying comparison between facility carbon per dollar and last-mile carbon per dollar unchanged. The representative inbound distance isolates the proximity effect of adding DCs. If density also changes inbound routing, shipment consolidation, or facility scale economies, those mechanisms add centralizing forces and can move a near-boundary case toward consolidation. The base model keeps last-mile rates fill-independent; the load-factor sensitivity of Section 4 shows how this assumption matters: common fill penalties consolidate both optima, while fill penalties that fall more heavily on carbon can reverse the density comparison. Service enters as a cycle-service requirement; if service or fill rate is chosen endogenously, safety stock becomes an additional design margin and the speed threshold should be evaluated at the

chosen service level. Finally, the calibration is regime-representative rather than firm-estimated, so the numerical values illustrate the order of magnitude of the threshold movements under documented rate ranges. The thresholds are therefore diagnostics to be evaluated with product- and network-specific rate data.

Future research. The most direct extensions are those that can move the threshold boundaries. Multi-product and multi-echelon settings would show how a shared network accommodates products in different regimes and where safety stock should be held. Explicit load-factor, shipment-consolidation, and routing models would quantify the centralizing force represented here by the load sensitivity. Heterogeneous geography and discrete sites would replace the continuous density curve with network-specific last-mile profiles. Empirical calibration with firm-level logistics-emissions data would estimate where real products fall relative to the speed and density boundaries, and endogenous service choice would make the cycle-service target a design decision rather than an input.

The model identifies which operational activity supplies availability and which supplies proximity at lower carbon intensity per dollar, and shows how the two substitutions interact in the joint distribution design: the speed threshold prices the availability margin, the pooling-adjusted density threshold prices the proximity margin, and complementarity links the two in the final configuration.

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